

1. (a) Provide definitions for the following terms:

- Normal form game.
- Strictly dominated strategy.
- Weakly dominated strategy.
- Best response strategy.
- Mixed strategy Nash equilibrium.

[5]

(b) State and prove a theorem giving a condition for which a strategy of the row player is a best response to a given strategy of the column player. [8]

(c) Consider the following Normal Form Game defined by:

$$M_r = \begin{pmatrix} 4 & -2 \\ -1 & 3 \end{pmatrix} \quad M_c = \begin{pmatrix} 2 & -2 \\ -3 & 2 \end{pmatrix}$$

State and justify which pairs of strategies are best responses to each other:

- (i) $\sigma_r = (1, 0)$ and $\sigma_c = (0, 1)$
- (ii) $\sigma_r = (1/5, 4/5)$ and $\sigma_c = (0, 1)$
- (iii) $\sigma_r = (5/9, 4/9)$ and $\sigma_c = (1/2, 1/2)$

[8]

(d) Using your answer to (c) or otherwise, find all Nash equilibria for the game. [4]

2. Consider the donation game defined by:

$$M_r = \begin{pmatrix} b+c & c \\ b+2c & 2c \end{pmatrix} \quad M_c = \begin{pmatrix} b+c & b+2c \\ c & 2c \end{pmatrix}$$

- (a) Show that if $b > c > 0$ then this game is a Prisoner's Dilemma. [3]
- (b) Obtain all Nash equilibrium for this game assuming the constraints of (a). [2]
- (c) Consider a Moran Process on this game. Obtain an expression for the fixation probability of i mutants: playing the first strategy in a population of with N as a function of b, c and N .

You may use the following expression for the fixation probability in the general two type Moran process:

$$\rho_i = \frac{1 + \sum_{j=1}^{i-1} \prod_{k=1}^j \gamma_k}{1 + \sum_{j=1}^{N-1} \prod_{k=1}^j \gamma_k}$$

where:

$$\gamma_k = \frac{f_R(i)}{f_M(i)}$$

Where $f_R(i)$ and $f_M(i)$ is the fitness of a resident/mutant respectively in a population with i mutants. [8]

- (d) Obtain the probability of a single mutant taking over for $N \in \{2, 3, 4\}$. [6]
- (e) For $N = 4$ consider the limit as $b \rightarrow \infty$ and as $b \rightarrow c$. Comment on the implications of these results. [6]

3. (a) Define a characteristic function game $G = (N, v)$.

[2]

- (b) Define the Shapley value.

[2]

- (c) Obtain the Shapley value for the following characteristic function games:

$$v_1(c) = \begin{cases} 0, & \text{if } c = \emptyset \\ 8, & \text{if } c = \{1\} \\ 5, & \text{if } c = \{2\} \\ 9, & \text{if } c = \{3\} \\ 10, & \text{if } c = \{1, 2\} \\ 11, & \text{if } c = \{2, 3\} \\ 18, & \text{if } c = \{1, 3\} \\ 30, & \text{if } c = \{1, 2, 3\} \end{cases} \quad v_2(c) = \begin{cases} 0, & \text{if } c = \emptyset \\ 80, & \text{if } c = \{1\} \\ 10, & \text{if } c = \{2\} \\ 12, & \text{if } c = \{3\} \\ 80, & \text{if } c = \{1, 2\} \\ 12, & \text{if } c = \{2, 3\} \\ 80, & \text{if } c = \{1, 3\} \\ 80, & \text{if } c = \{1, 2, 3\} \end{cases}$$

[8]

- (d) Given a game $G = (N, v)$, a payoff vector λ satisfies the symmetry property if, for any pair of players i, j :

If $v(C \cup i) = v(C \cup j)$ for all coalitions $C \subseteq \Omega \setminus \{i, j\}$, then:

$$\lambda_i = \lambda_j$$

Prove that the Shapley value satisfies the symmetry property.

[6]

- (e) The additivity property is:

Given two games $G_1 = (N, v_1)$ and $G_2 = (N, v_2)$, define their sum $G^+ = (N, v^+)$ by:

$$v^+(C) = v_1(C) + v_2(C) \quad \text{for all } C \subseteq \Omega$$

A payoff vector λ satisfies the additivity property if:

$$\lambda_i^{(G^+)} = \lambda_i^{(G_1)} + \lambda_i^{(G_2)}$$

Using the two games from part (c), demonstrate that the Shapley value has the additivity property.

[7]

4. (a) Define a **social welfare function**.

[2]

- (b) State **Arrow's Impossibility Theorem**. Briefly discuss the implications of this theorem and ways in which the **Borda** or **Condorcet** methods respond to this impossibility.

[5]

- (c) Consider the following preference profile over the set of alternatives $X = \{A, B, C\}$:

Number of voters	1st choice	2nd choice	3rd choice
4	A	B	C
3	B	C	A
2	C	A	B

- (i) Construct the pairwise majority contests among the three alternatives.
(ii) Determine if there is a **Condorcet winner**.
(iii) Explain whether the collective preference relation using this approach is transitive.

[8]

- (d) Apply the **Borda count** method to the same profile.

- (i) Compute the Borda scores for each alternative.
(ii) Identify the Borda winner.
(iii) Does the Borda method select the same outcome as the Condorcet method?

[7]

- (e) Define what it means for a voting rule to satisfy the **Independence of Irrelevant Alternatives (IIA)** property. Then, using the Borda count, give an example or explanation of how IIA may fail.

[3]